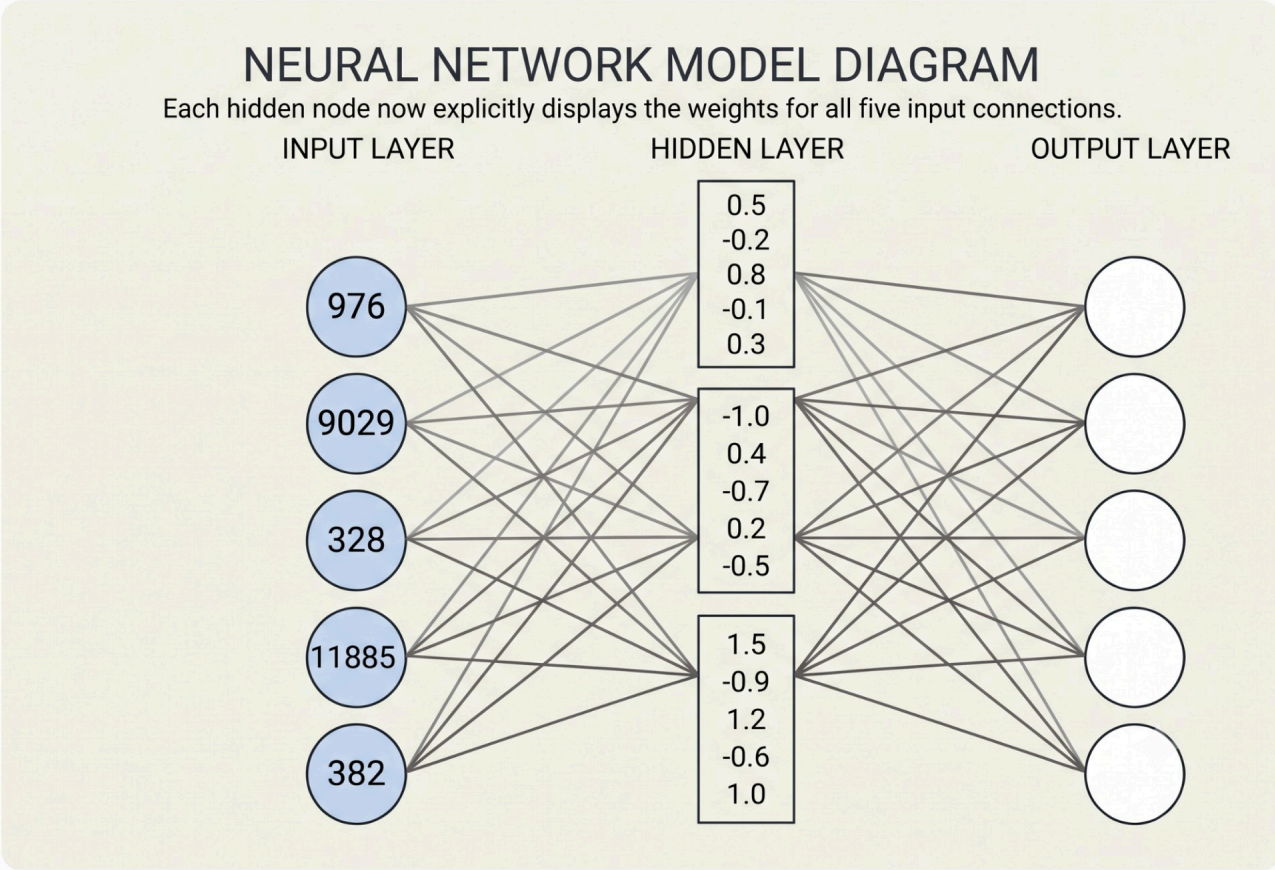
But how does the LLM *know* what word should come next?

The Magic Comes from the Weights

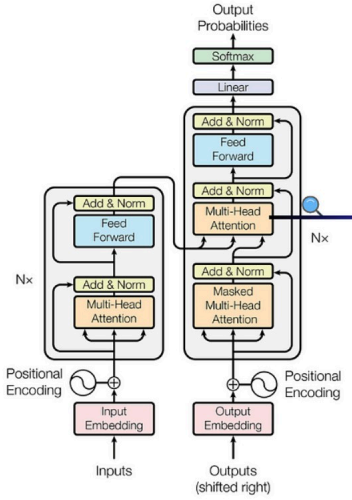
Our simple network: **15 parameters**.



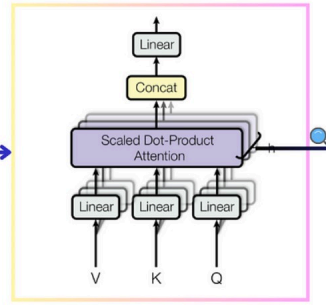
Modern LLMs

1 trillion+

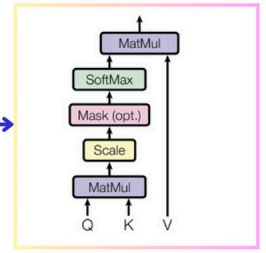
parameters



Transformers



Muti-Head Attention



Scaled Dot-Product Attention

Picking Good Weights

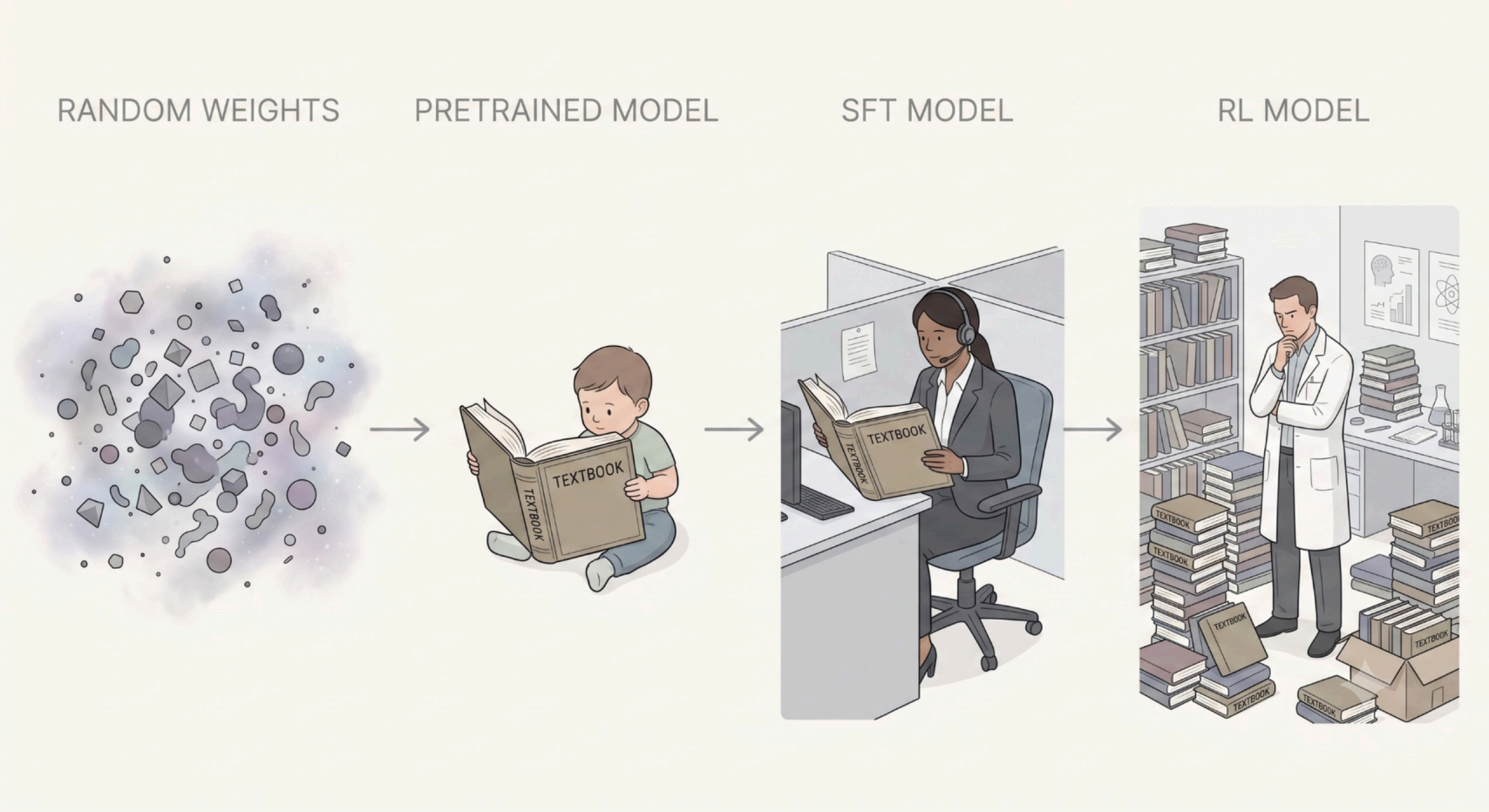
Initially, the weights are just **random numbers**.

We have to **train** the model to pick good weights.

Good weights mean good predictions.

Training Language Models

The Training Process

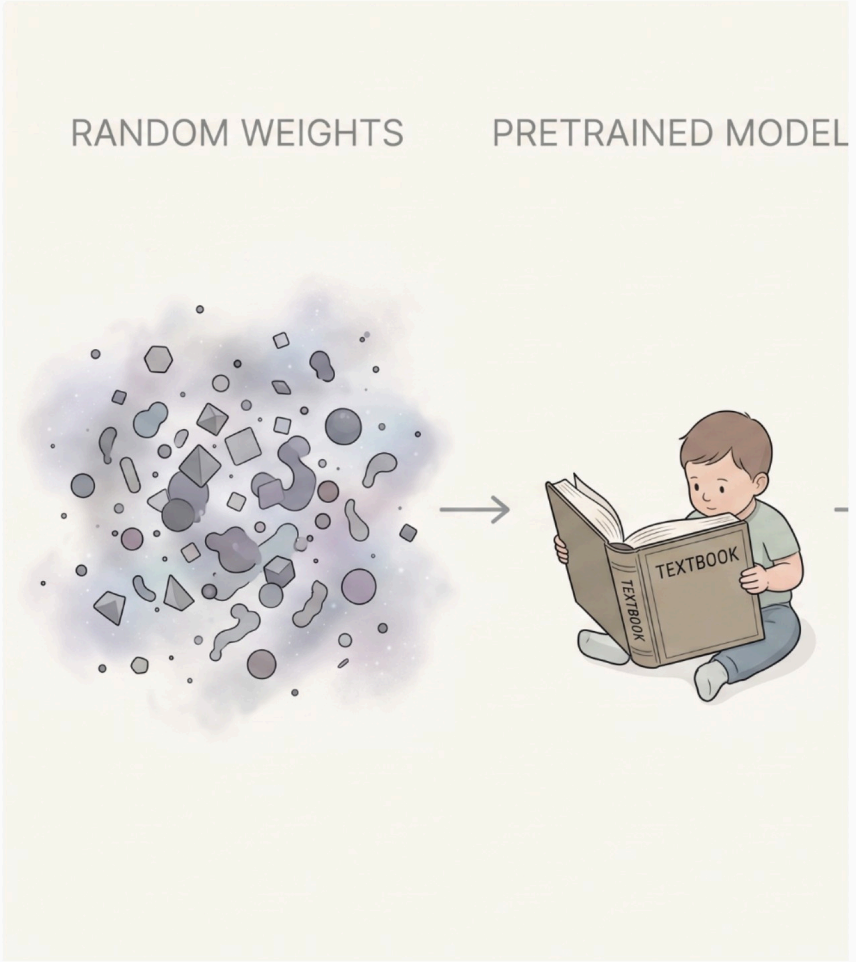


1. Pretraining

15–20 trillion

words of text used to train GPT-5

Training: see a snippet of text, **predict the next word**.



Step 1: Guess

The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

Untrained model's top guesses

`keen` `íf` `PN` `much` `wild`

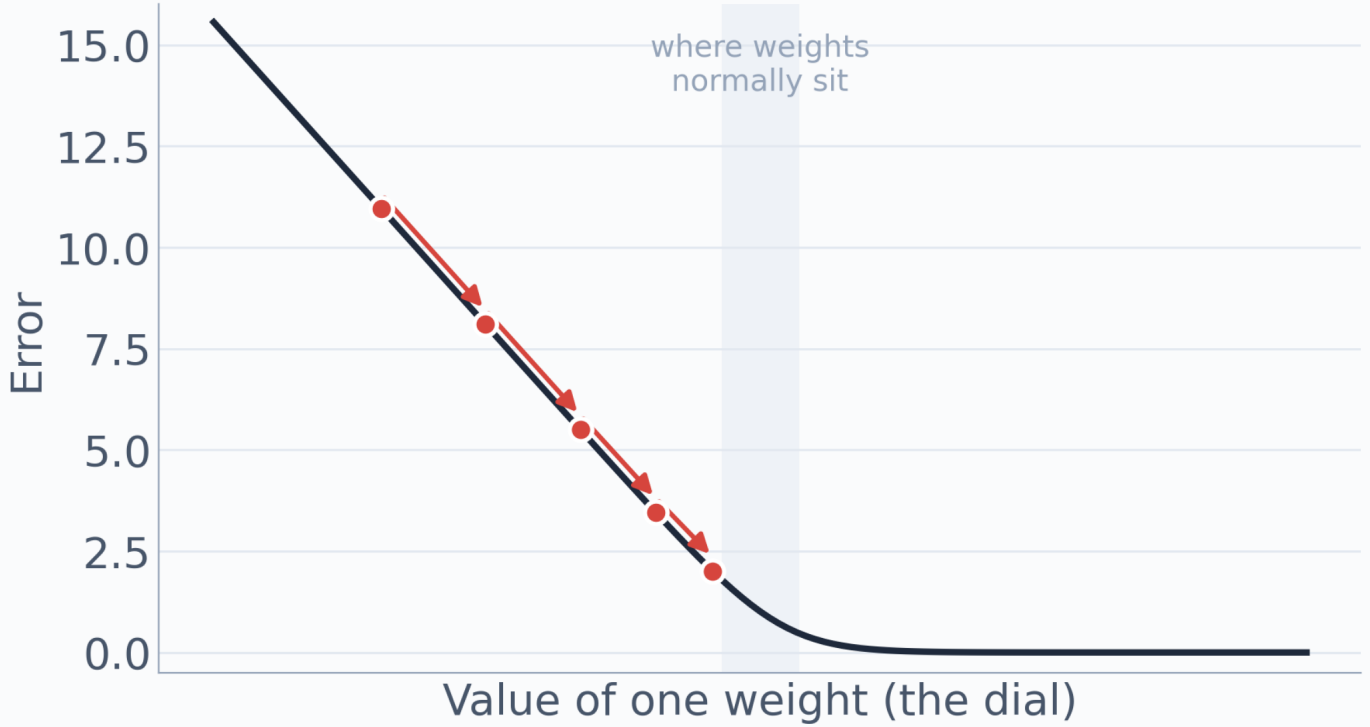
"Austin": **0.0015%**

Step 2: Measure the Error

Checkpoint	P("Austin")	Error
Step 0	**0.0015%**	11.1
Step 2,000	**0.04%**	7.8
Step 8,000	**17%**	1.8
Step 143,000	**36%**	1.0
Perfect	**100%**	0

Step 3: Turn the Dial Downhill

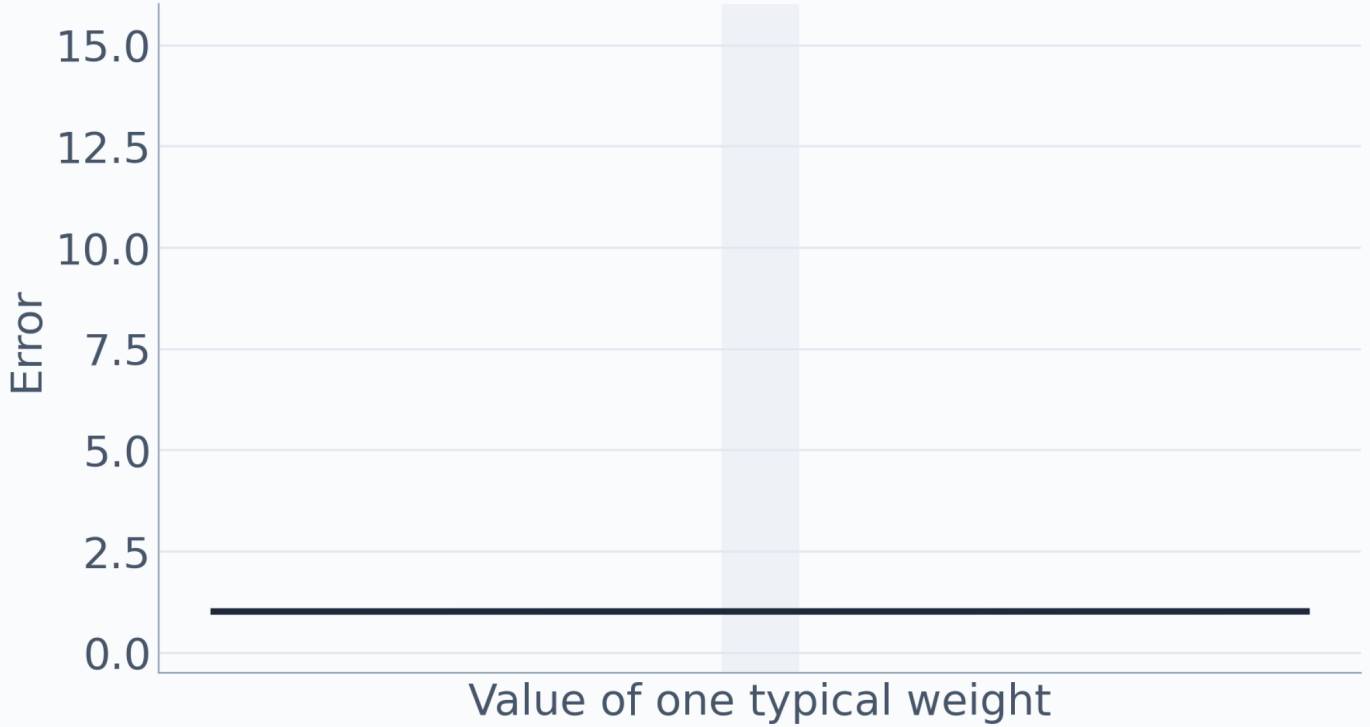
One weight = **one dial**.



But Most Single Dials Barely Matter

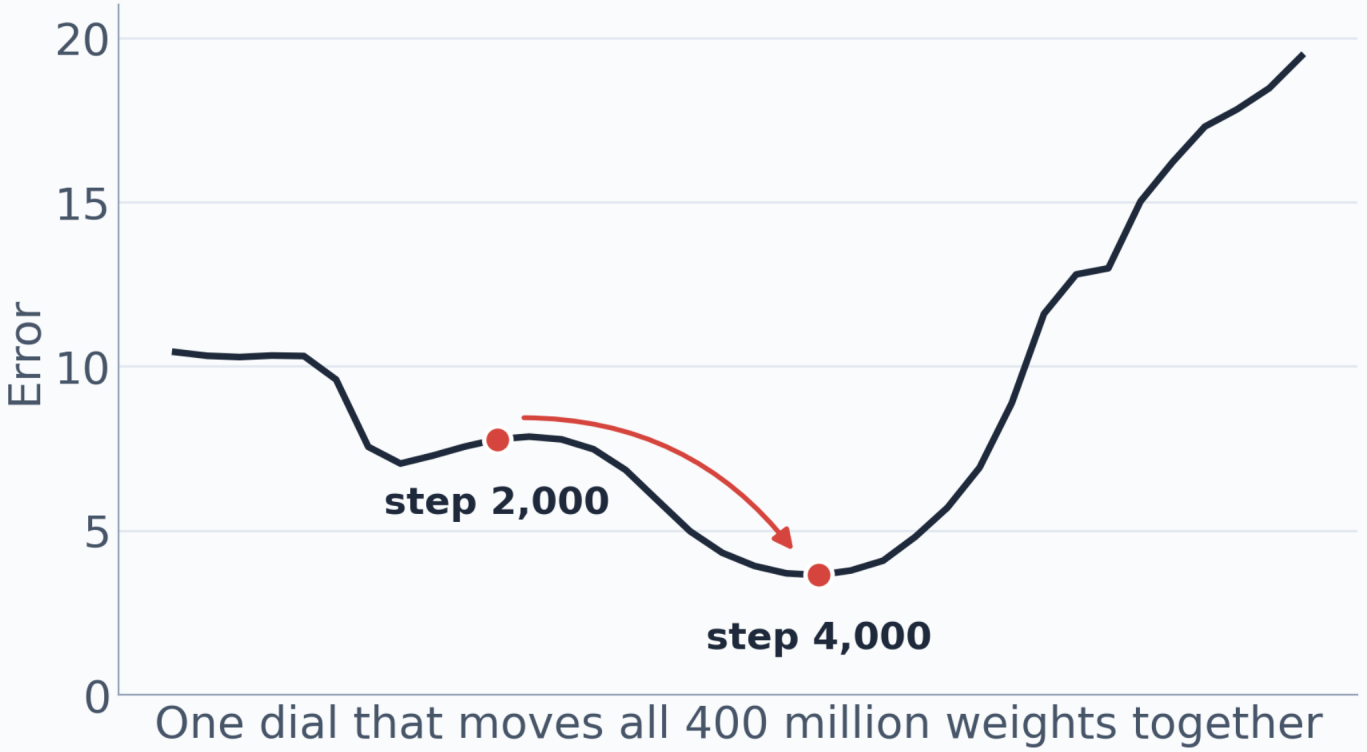
400 million

weights in this model



So Training Turns Every Dial at Once

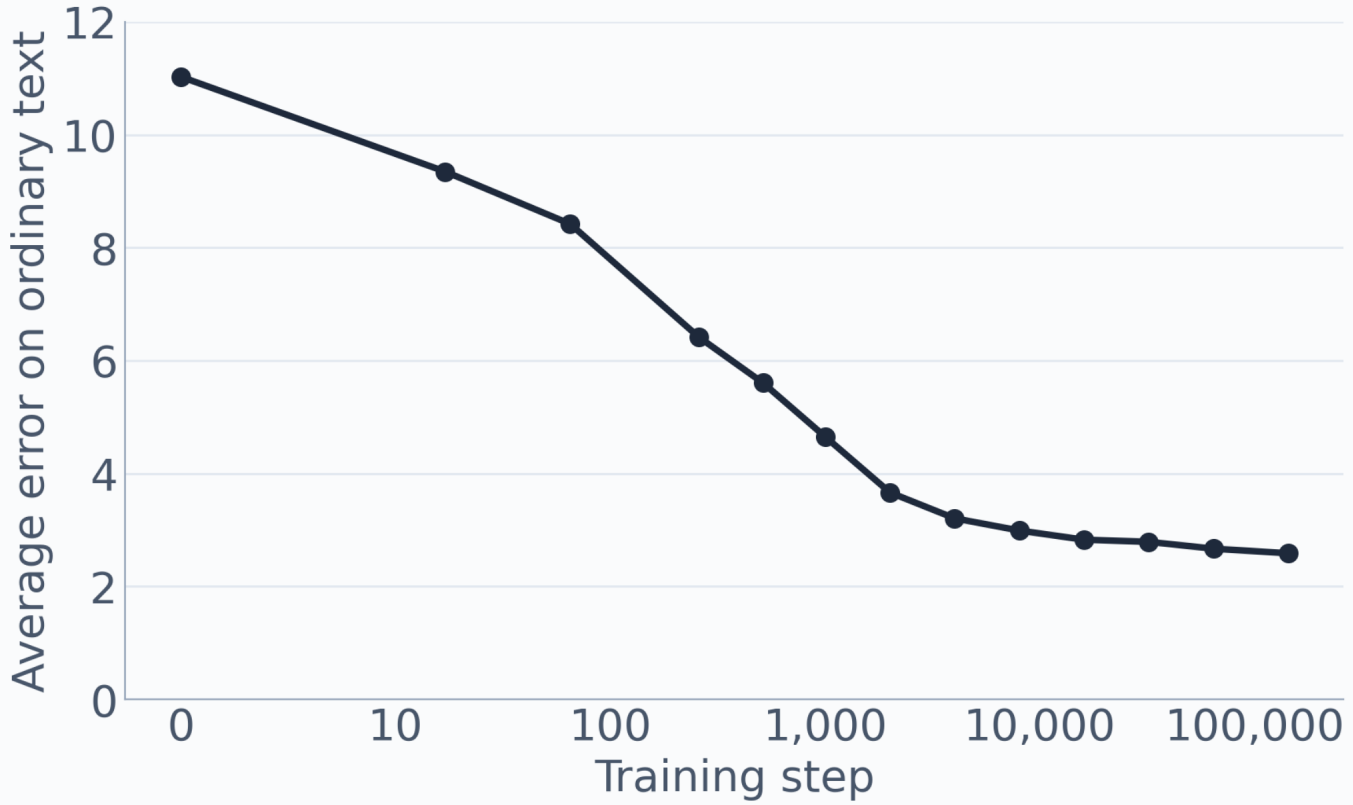
Each step nudges **all** the weights a tiny amount.



Then Repeat, 143,000 Times

143,000
training steps

300 billion
tokens of text



Same Shape at Frontier Scale

2 trillion

tokens of text

DeepSeek LLM 7B and 67B (DeepSeek-AI, 2023)

